AXIOM / MMF

Geometry, derived from first principles.

COMPUTATIONAL DESIGN · CAD · AUTOMATION · FABRICATION

CAPABILITY STATEMENT

A DIVISION OF
MILFORD MACHINING & FABRICATION
PRINCE EDWARD COUNTY, ON, CANADA

Axiom designs parts and systems as code: rule-driven, voxel-based geometry delivered as native Rhino files, and built by hands that machine metal. The computational design division of Milford Machining & Fabrication (MMF), backed by 10+ years of CAD, machining, industrial machinery, machine safety and automation controls.

01 CORE COMPETENCIES

Computational design
Rule-driven and voxel geometry in Axiom, our in-house engine on the open-source PicoGK kernel.

Native .3dm delivery
Clean Rhino files your team can open, edit and fabricate from. No lock-in.

Industrial machinery
Design, fabrication, installation and commissioning, with machine safety in mind.

Automation & controls
AutomationDirect CLICK PLCs, Shelly and ESP32 monitoring, low-voltage panels; mains wiring by a licensed electrician.

CAD & drafting
Parametric models, assemblies, drawings to ASME Y14.5, as-built 3D capture.

02 SERVICES

AXIOM **Computational design**
Feasibility studies, parametric part families, lattices, conformal channels and AM-ready geometry, with a written summary of the rules and inputs behind each output.
`Rhino .3dm · STL / 3MF · STEP (where geometry allows)`

CAD **CAD for hire**
Constrained parametric part models and assemblies, 2D production and GA drawings, PDF or sketch to native CAD, design-intent notes.
`STEP · IGES · SolidWorks · Rhino .3dm · DWG/DXF · PDF`

CONTROLS **Automation**
Commented programs, labelled I/O, wiring and panel drawings, hand-over notes; commissioning on site in Ontario or remote.
`CLICK project files · ESPHome/Arduino source · PDF · DWG`

MMF **Machining & fabrication**
Machined parts, fixtures, welded assemblies and machine design-and-build; prototype and short-run, with DFM feedback.
`Quoted from STEP, DWG/DXF or PDF drawings`

SOFTWARE **Rhino voxel-to-.3dm plugin**
Voxel fields to clean, native Rhino geometry. Planned for food4Rhino; early-access list open.

03 DIFFERENTIATORS

Rules, not redraws. Change a requirement, regenerate the design. Every output traceable to its inputs.

Design and build under one roof. One team from drawing to finished part: Axiom designs it, MMF makes it.

Buildable by default. Every rule is checked against how the part will be cut, welded, wired and maintained.

You own it. Native files, documented rules, labelled wiring. Deliverables are yours on full payment.

Plain terms. Fixed quotes after scoping; deliverables, revisions and timeline agreed in writing.

Export-control aware. Controlled-goods, ITAR, defence and rocket work is reviewed before any files are exchanged.

04 EQUIPMENT & TOOLS

MACHINING	WELDING	DESIGN
CNC lathe	TIG	Rhino 8 (CAD)
CNC mill	MIG	PicoGK / Axiom computational stack
CNC plasma table	Stick	

05 PAST PERFORMANCE

A CROSS-SECTION OF PAST PROJECTS

MACHINERY
Gantry positioning system
Multi-axis gantry, designed, built and commissioned.

MACHINERY
Box flipper & conveyors
Custom flipper with matched infeed and outfeed conveyors.

CONTROLS
Vacuum control panels
Industrial vacuum panels rebuilt, rewired and documented.

CONTROLS
Tank autofill monitor
Level monitoring and automatic fill for an industrial tank.

AEROSPACE
Long-endurance UAV
An uncrewed aircraft design project.

ENERGY
Off-grid security container
Solar-powered monitoring, Starlink-connected.

ENERGY
Solar command trailer
Mobile command station, deployable solar panels.

ENERGY
Dual-axis solar tracker
Solar array that follows the sun on two axes.

ARCHITECTURE
Architectural elements
Fabricated features, from model to installed piece.

AND MORE
Fixtures & retrofits
Repairs, retrofits and one-off machines.

FLAGSHIP · AEROSPACE / PROPULSION

RDE-2: methalox rotating detonation engine

Designed end to end in Axiom: fully computational geometry from a single-source rule set, exported as a native .3dm model and designed for additive manufacturing.

Technical details are withheld. Propulsion data may be subject to Canadian and US export controls.

06 COMPANY DATA

BUSINESS NAME Milford Machining & Fabrication	DIVISION Axiom, computational design
BUSINESS TYPE Sole proprietorship	REGISTERED May 2023, Ontario
LOCATION Prince Edward County, ON	COUNTRY Canada

Bring us the hard part.

One intake form for requirements, files and timeline. Reply within two business days.

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WEB
axiom-mmf.ca

PROJECT INTAKE
axiom-mmf.ca/intake.html